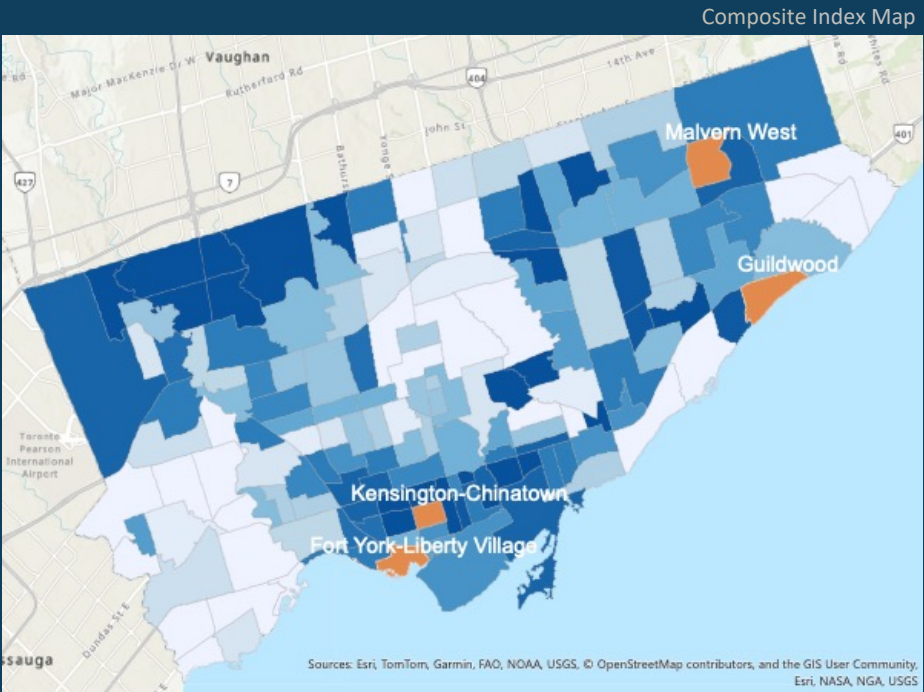
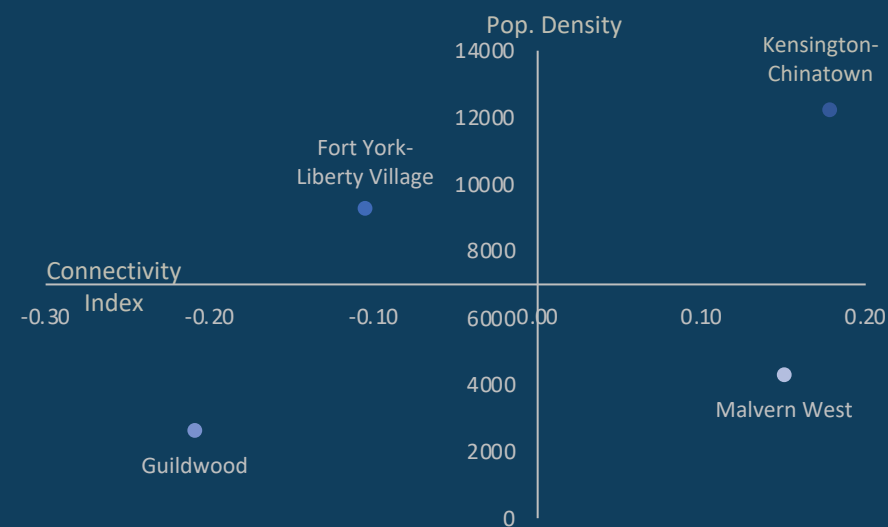


Measuring Neighbourhood Social Connectivity

Five-indicator Composite Index

This sample of my data analysis and GIS work is adapted from a study proposal measuring social connectivity in four Toronto neighbourhoods. It is also available as an [ArcGIS Story Map](#).



Methodology

1. Five indicators of social connectivity were selected from the Statistics Canada Neighbourhood Profiles dataset:

- Structural Type of Dwelling
- Household Type
- Household Size
- Mode of Commuting
- Commuting Time

Together, these indicators capture key demographic and mobility characteristics associated with social isolation or connection.

2. Responses were classified into "Low" and "High", with the difference scores between the mean values of each group aggregated into a composite index.

3. The dataset was cleaned and formatted as an attribute table, then joined with spatial data available from the City of Toronto Open Data Portal. Choropleth maps were produced visualizing the spatial distribution of the indicators and composite index.

Key Findings

This composite index captures a multi-dimensional view of social connectivity that complements population density while remaining independent, suggesting that **highly active social environments exist in both high-density, walkable downtown areas and in some low-density suburban neighbourhoods**. The distribution also reveals several interesting outliers. Four neighbourhoods were selected for further study.